VIPP Report

Name: **Youssef Al Hajj Youness**

AUB ID: **202407879**

American University of Beirut

Introduction

In this report, I evaluated the `prostate_mri_anatomy` model from MONAI for prostate MRI segmentation. A description of the model can be found at this MONAI page. I used the `PROMISE12_nifti` dataset, which is available at the PROMISE12 challenge website. The dataset consists of **T2-weighted prostate MRI scans** with corresponding **whole-prostate segmentation masks** stored in NIfTI format. Therefore, the evaluation was carried out as a whole-gland segmentation task. At the end of the evaluation, I obtained an average **whole-gland Dice score of 0.8504**.

Qualitative Comparison in 3D Slicer

The following screenshots were taken from 3D Slicer to visually compare the provided ground-truth annotation and the model prediction. In these screenshots, the **annotation is shown in green**, while the **model output is shown in red**.

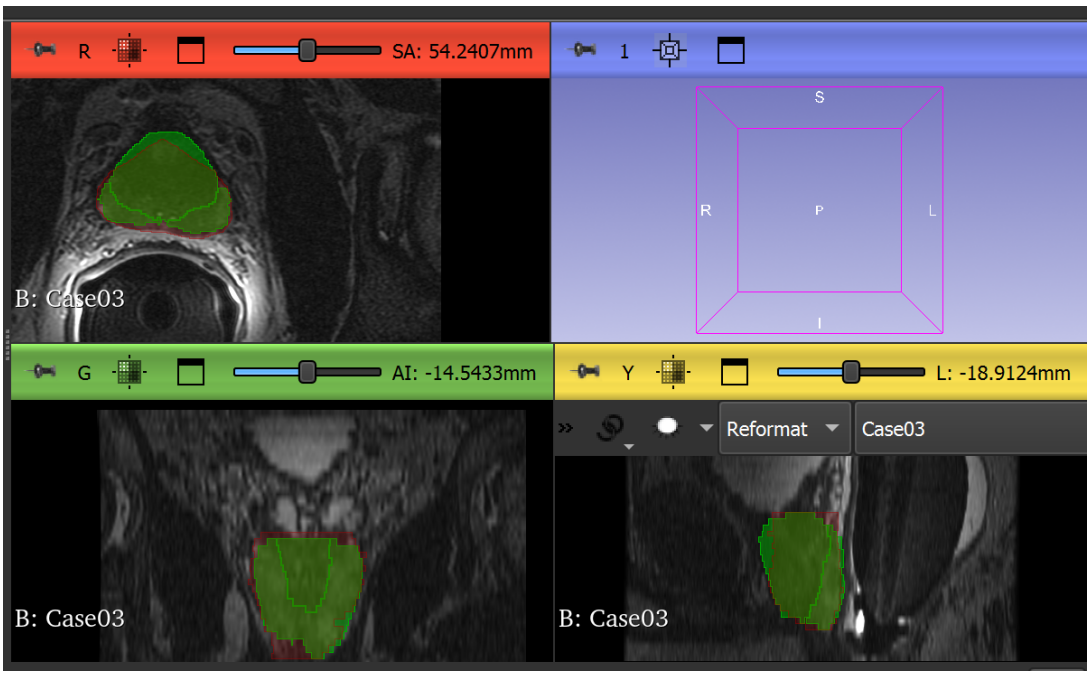

Figure 1: 3D Slicer comparison between the ground-truth annotation (green) and the model output (red).

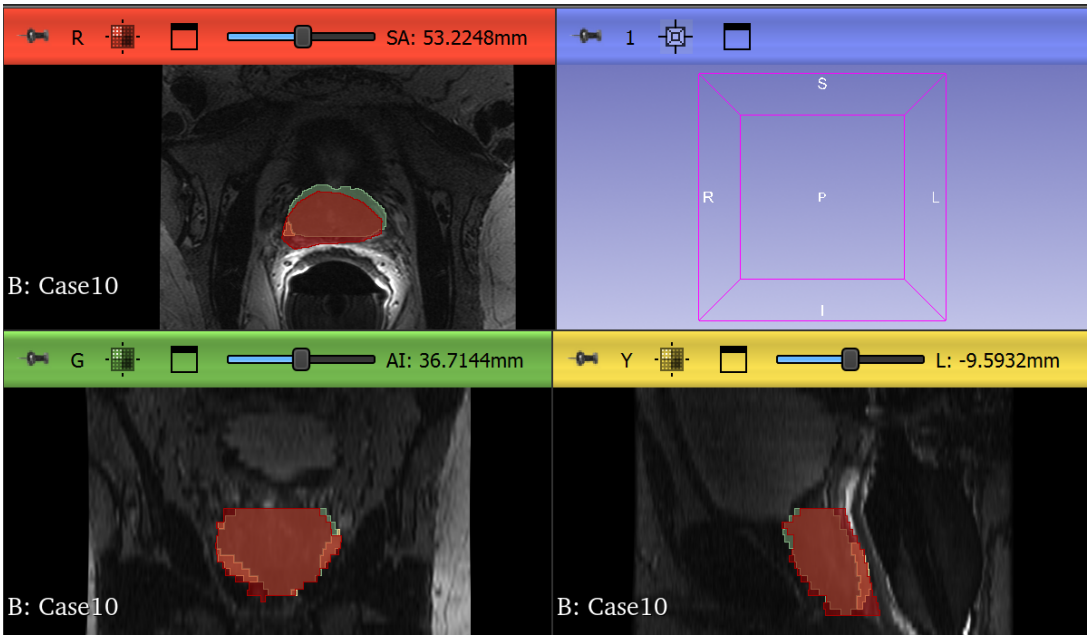

Figure 2: 3D Slicer comparison between the ground-truth annotation (green) and the model output (red).

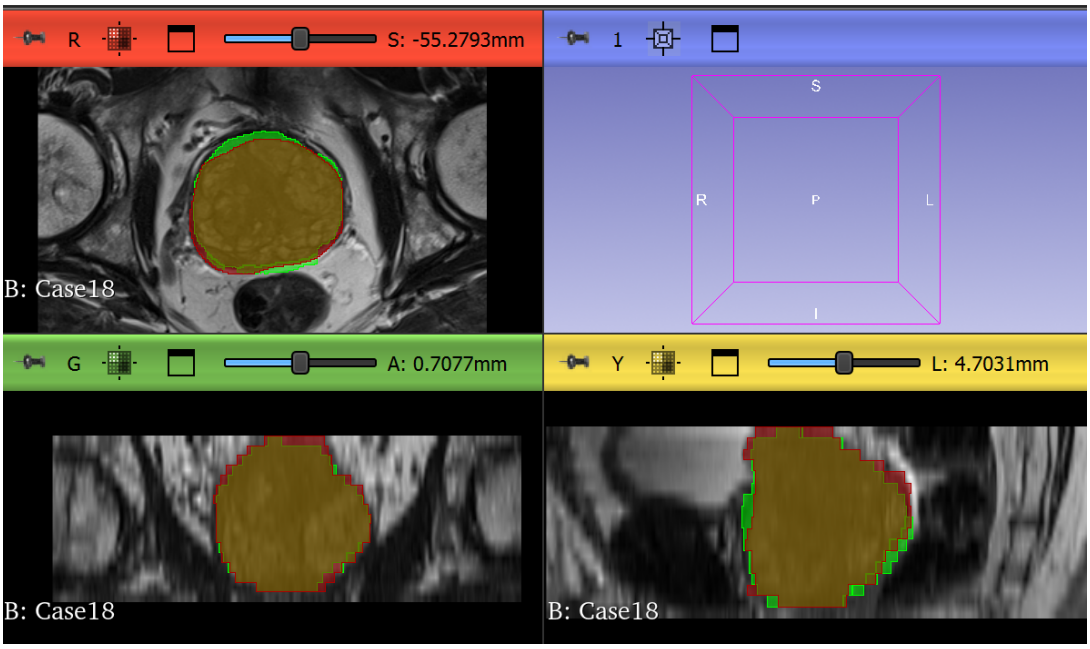

Figure 3: 3D Slicer comparison between the ground-truth annotation (green) and the model output (red).

Quantitative Results

Case	Mode	Whole Gland Dice
Case00	whole-gland binary GT	0.8714
Case01	whole-gland binary GT	0.8728
Case02	whole-gland binary GT	0.8244
Case03	whole-gland binary GT	0.8744
Case04	whole-gland binary GT	0.7975
Case05	whole-gland binary GT	0.8524
Case06	whole-gland binary GT	0.8415
Case07	whole-gland binary GT	0.7977
Case08	whole-gland binary GT	0.8602
Case09	whole-gland binary GT	0.8572
Case10	whole-gland binary GT	0.8446
Case11	whole-gland binary GT	0.8914
Case12	whole-gland binary GT	0.8551
Case13	whole-gland binary GT	0.8710
Case14	whole-gland binary GT	0.9202
Case15	whole-gland binary GT	0.9038
Case16	whole-gland binary GT	0.8996
Case17	whole-gland binary GT	0.2439
Case18	whole-gland binary GT	0.9236
Case19	whole-gland binary GT	0.8898
Case20	whole-gland binary GT	0.9000
Case21	whole-gland binary GT	0.8880
Case22	whole-gland binary GT	0.9121
Case23	whole-gland binary GT	0.9378
Case24	whole-gland binary GT	0.9288
Dataset mean	whole-gland binary GT	0.8504

Table 1: Dice score results obtained on the `PROMISE12_nifti` dataset. Since the ground truth consists of binary whole-gland annotations, zonal Dice scores are not applicable.